

Davidious AI Code Lab Trainer

Project Topic (Tema)

AI Trainer for Coding, Tracking and Evaluating progress.

Description

AI coding lab with task generation, sandbox code execution, evaluation, progress tracking, and timed interview simulation.

Requirements and Specifications

- AI-driven coding task generation
- Dockerized secure sandbox code execution
- Automated solution evaluation system
- Timed interview simulation module
- Progress tracking and analytics
- Documents and youtube course integration
- Email reminders to practise weak skills

Programming languages

Python(LangChain+OpenAI) +FastAPI(SQLModel ORM) +React +Typescript

AI Agents

1.Coding Task Generation Agent

- Generates coding challenges based on skill and difficulty .

2. Evaluation Agent

- Analyzes user solutions, scores them, and provides structured feedback.

3. Learning Strategy Agent

- next skill focus
- difficulty level
- when to repeat topics
- when to introduce new concepts

4. Interview Simulation Agent

- Conducts timed challenges and evaluates under interview conditions.

Programming languages

Python(LangChain+OpenAI) +FastAPI(SQLModel ORM) +React +Typescript